

API and Data Source Reference Guide

This document provides a consolidated reference for all external APIs and data sources required for the "Global Firefight: Drone Command" project.

1. NASA FIRMS (Fire Information for Resource Management System)

Provides near real-time, global active fire hotspot data. This is the primary source for dynamic in-game events.

- **Primary Purpose:** To populate the game world with active fire locations, which serve as the main gameplay objective.
- **Authentication:** Requires an Earthdata Login API key.
 - **Registration URL:** <https://firms.modaps.eosdis.nasa.gov/web-services/>
- **Recommended Endpoint (Active Fire Data):** Downloadable CSV of active fire data. This provides the most detailed information for gameplay mechanics.
 - **Data Format:** CSV
 - **Base URL for Download:** https://firms.modaps.eosdis.nasa.gov/api/active_fire/
 - **Example Request URL (VIIRS, World, 24 hours):**
https://firms.modaps.eosdis.nasa.gov/api/active_fire/c1/csv/YOUR_API_KEY/VIIRS_SNP_P_NRT/world/24h
- **Key Data Fields to Use:**
 - latitude, longitude: For placing the fire on the globe.
 - confidence: To determine the fire's initial "health".
 - frp (Fire Radiative Power): To determine the fire's "intensity" and spread rate.

2. NASA EONET (Earth Observatory Natural Event Tracker)

Provides curated data on major, named natural events, including significant wildfires. Used to add narrative context and high-priority missions.

- **Primary Purpose:** To identify major wildfire events and create high-value "mission" objectives in the game.
- **Authentication:** No API key is required for public access.
- **Recommended Endpoint (Active Wildfire Events):**
 - **Data Format:** GeoJSON
 - **Endpoint URL:**
<https://eonet.gsfc.nasa.gov/api/v3/events?category=wildfires&status=open>
- **Key Data Fields to Use:**

- title: The human-readable name of the fire (e.g., "Kincade Fire").
- geometry: The coordinates (Point or Polygon) for placing a mission marker on the globe.
- link: A URL to a NASA article about the event for optional in-game information panels.

3. Google Maps Platform (via Cesium for Unity)

Provides the high-resolution, photorealistic 3D world model for the tactical, third-person drone view.

- **Primary Purpose:** To serve as the high-fidelity visual backdrop for close-up gameplay.
- **Authentication:** Requires a Google Cloud API key with the "Map Tiles API" enabled and billing configured.
 - **Setup Console:** <https://console.cloud.google.com/>
- **Service:** Map Tiles API - Photorealistic 3D Tiles
- **Endpoint URL (for direct connection in Cesium for Unity):**
 - **Data Format:** 3D Tiles (consumed by Cesium)
 - **Endpoint URL:**
`https://tile.googleapis.com/v1/3dtiles/root.json?key=YOUR_API_KEY`
- **Critical Requirement:** You **must** display Google's data attribution on screen. In the Cesium3DTiles component in Unity, ensure the Show Credits On Screen property is checked true.